# Descriptive Statistics

* Types of Data Analysis :-

(1) Visual & Non-Visual Analysis

↓ analyze the data using graphs & charts

↳ analyze the data using nos., calculations, etc. instead of graphs.

(2) Univariate & Bivariate Analysis

↓ uni → one

Here we study one column at a time

↳ Bi → two

Here, we study two columns together

(3) Descriptive Analysis

↓ Tells us that what has happened in the data

Common statistics of Descriptive Analysis :-

a) Mean
b) Mode
c) Median
d) Standard Deviation
e) Variance
f) Range
g) Percentiles
h) Frequency

(4) Inferential Analysis

Used to make conclusion about the entire population

based on a sample.

⑤ Probabilistic Analysis

↳ Used to analyze the data based on the probability

↳ Used to check how likely is an event to occur.

* Statistics :-

It is a branch of mathematics that involve the collection, analysis, interpretation, etc. about the data

* Descriptive Statistics :-

It focus on summarizing raw data to make it understandable

↓

How??

↓

Using various factors / parameters ⇝ measures of central tendency
↳ measures of dispersion, etc.

* Why statistics are important for data science?

↓

- It is the backbone of understanding & interpreting the data.
- It plays an important role across fields like ML, DL, GenAI

* What is data?

It refers to a piece of information where we process the facts and figures about it.

* Types of data :-

① Categorical data → ex- Name, City, Qualifications, etc.

(1) Categorical data → ex- Name, City, Qualifications, etc.

(2) Nominal data → Type of a categorical data where categories differ from each other.

ex- Employee ID, Telephone No., etc.

(3) Ordinal Data → It consists of groups or categories which follows an order.

Ex- Qualification, Rank in a dataset

(4) Numerical Data → Consists of nos.

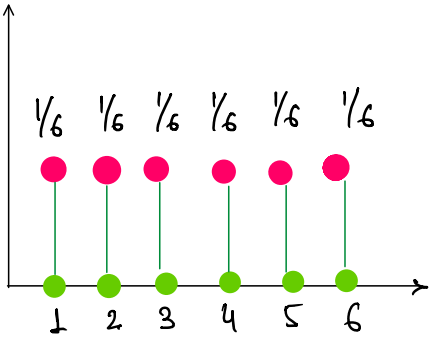
Data is divided broadly into two major categories → Discrete, Continuous

★ Discrete Data :-

↓

is defined at a specific interval time

Ex-



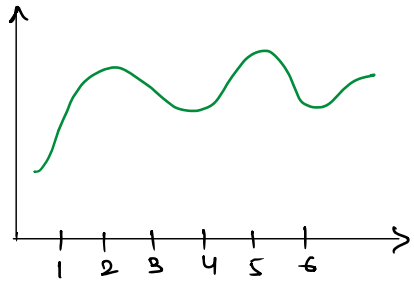

Die = 6 faces

Discrete Data → based on both numerical & categorical variables

★ Continuous Data :-

★ Continuous Data :-

Data is defined at each interval of time




Continuous Data → It is based on purely numeric data

★ Measures of Central Tendency:-

1. Mean
2. Median
3. Mode

Mean :- It is the average value.

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Total no. of observations}}$$

Ex- 2, 5, 8, 3, 7, 9

$$\text{Mean / Avg} = \frac{2+5+8+3+7+9}{6}$$

$$\bar{X} = \frac{34}{6} = 5.66$$

Median :- It is the middle value of the sorted data.

Ex- 2, 5, 8, 3, 7, 9

Sort → 2, 3, 5, 7, 8, 9

Here, $n = 6$ (even)

$$\text{Median} = \frac{5+7}{2} \longrightarrow [\text{sum of two middle values}]$$

$$= 6$$

In case of even :-

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{th} + \left(\frac{n}{2}+1\right)^{th}}{2}$$

$$= \frac{3^{rd} + 4^{th}}{2}$$

$$= \frac{5+7}{2} = 6$$

Ex (2) → 4, 3, 5, 1, 8

Sort → 1, 3, 4, 5, 8

↓

Middle value

So, Median = 4

When $n$ is odd, $n = 5$

$$\text{Median} = \left(\frac{n+1}{2}\right)^{th} \text{ term}$$

$$= \left(\frac{5+1}{2}\right)^{th} \text{ term}$$

$$= 3^{rd} \text{ term}$$

Median = 4

Mode :- The value occuring more no. of times

↳ Highest frequency value

Ex- (2), 4, 1, 4, (2), 3, 6, 8, 1, (2), 5, 7, 9, (2), 6

Here, Mode = 2

* Measures of Dispersion :-

1. Range
2. Variance
3. Standard Deviation
4. Percentiles

Range :- The difference b/w the Maximum value & minimum value.

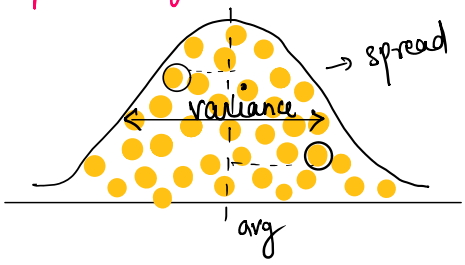
17, 10, 20, 40, 25, 18, 30, 45, 50, 42

min = 10 , max = 50

Range = max - min

= 50 - 10 = 40

Variance :- Spread of the data




$$Var = \frac{\sum_{i=1}^{N} (X_i - \bar{X})^2}{N}$$

$$= \frac{(X_1 - \bar{X})^2 + (X_2 - \bar{X})^2 + \text{----}}{N}$$

| A | B |
|---|---|
| $X_1$ | $Y_1$ |
| $X_2$ | $Y_2$ |
| $X_3$ | $Y_3$ |
| ⋮ | ⋮ |

Avg → $\bar{X}$ Avg → $\bar{Y}$

where, $X_i$ = individual datapoint

$\bar{X}$ = mean / avg

N = Total values in the column (length)

Standard Deviation :- It also explain the spread of the data.

↓

Square Root of Variance

$std = \sqrt{var}$ ⇝ If we already calculate variance then why std?

Ans → $X_1 = 2$

$X_2 = 6$ $\bar{X} = 4$

$X_3 = 5$

⋮

(Assume without square) $var = \frac{(X_1 - \bar{X}) + (X_2 - \bar{X}) + \text{----}}{N}$

(Assume without square) $Var = \frac{(X_1-\bar{X}) + (X_2-\bar{X}) + \cdots}{N}$

$= \frac{(2-4) + (6-4) + \cdots}{N}$

$= \frac{-2 + 2 + \cdots}{N}$

Due to the cancellation, we cannot capture the spread of $X_1$ & $X_2$

So, to avoid cancellation we take square

$= \frac{(-2)^2 + (2)^2 + \cdots}{N}$

different units are used ← we scale up the values ← $= \frac{4+4+\cdots}{N}$

↓

To make the result fall in the same unit ⇝ Std

**Percentage :-** This indicates the absolute score out of 100

Ex- Marks obtained by a student A is 95

So, Percentage $= \frac{95}{100} = 95\%$

**Percentile :-** It is the relative rank based on comparison with others.

Ex- Total students appearing in NEET = 1000

Student A scored = 688 marks

Total marks = 700

Percentage = 98%

but assume percentile → 94%

↓

what this means?

Out of 1000 students, 94% students scored marks below 688

Formula → $\text{No. of people} \leq \frac{\text{Your Score}}{\text{Total People}} \times 100$

Percentage depends only on our own correct answers
but
Percentile depends on our score and how well everyone else has performed